

MUHAMMAD HIDAYAH BIN MOHD LATIF

Software Engineer | AI Agent Systems & Full-Stack Development

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Available from July 2026 | BEng graduation Aug 2026

PROFESSIONAL SUMMARY

Software engineer specialising in AI agent systems and full-stack development. Built a production AI booking assistant from prototype to deployment at a healthcare logistics SaaS company — designing agent orchestration, multi-step tool workflows, and conversational state machines integrated with SAP enterprise systems. Contributed to open-source SAP tooling across multiple PRs. Currently building a React Native mobile platform for motorcycle owners. Seeking software engineering roles in AI applications, full-stack, or mobile development.

WORK EXPERIENCE

Software Development & Innovation Intern — mymediset (BIT Consultant Pte Ltd)

May 2025 – Present | Singapore · Hybrid | Medical device logistics SaaS platform with SAP backend

- Initiated and built the company's first AI prototype — a conversational booking assistant replacing multi-step SAP workflows with natural language, which became the reference architecture for the production integration.
- Designed a dual-agent system on Cloudflare Workers: 7-stage booking state machine, 20+ MCP tools across 7 categories, AI booking analysis with confidence scoring, and conversation memory compaction (~30–40% token savings).
- Achieved 5.5x latency improvement (1.2s → 220ms) via Cloudflare KV caching with stale-while-revalidate patterns.
- Built a Summary Agent for the production SAP CAP platform — dual LLM + deterministic classification for daily booking dashboards with attention severity rules and focus modes.
- Contributed to the open-source CAP MCP Plugin (SAP Cloud Application Programming + Model Context Protocol) — adding OData \$expand support and deep insert functionality across multiple PRs.
- Diagnosed and resolved API quota exhaustion during a live client demo; implemented Durable Object hibernation for WebSocket cost optimisation.
- Presented prototype to CEO and delivered demos to cross-functional multinational stakeholders.

Transport Operator — Singapore Army (National Service)

Oct 2020 – Oct 2022 | Singapore

- Operated Class 3 and 4 military vehicles for logistics and deployment operations. Promoted to Corporal First Class (CFC); mentored new operators and relieved sergeant duties for leadership continuity.

Engineering Intern — Denso Wave Singapore

Mar 2019 – Sep 2019 | Singapore

- Built and improved automated firmware test systems using Python and VBA, collaborating with hardware engineers to investigate and resolve issues. Offered contract extension for strong performance.

PROJECTS

Kickstand Mobile | React Native (Expo), NestJS, Drizzle ORM, Supabase, Mastra AI

2026 – Present

- Building a mobile-first platform for Singapore motorcycle owners — compliance tracking, maintenance logging, workshop price comparison, and an AI voice agent that understands your specific bike.
- Backend (NestJS): auth module with Supabase Auth proxy, bikes CRUD with ownership guards, mileage tracking with forward-only validation, 9-table Drizzle ORM schema, workshop proximity search via Haversine formula, and unit tests. Monorepo architecture.
- AI agent (planned): 9 bike-aware tools via Mastra framework, voice conversational interface, and proactive compliance notifications via background cron jobs.

AI Booking Chatbot — Capstone Project | SAP CAP, Cloudflare Workers, Claude API, MCP, React

2025 – 2026

- Architected a 4-stage evolution for medical appointment booking: Cloudflare Workers → MCP CDS Plugin → SAP CAP + MCP → SAP CAP + Native Tools. Core insight: MCP suits loosely-coupled entity access; native tools suit tightly-coupled

multi-step workflows.

- Integrated LLM tool-use patterns with SAP Cloud Application Programming Model for real-time booking against live healthcare data. Built 45+ React components for the conversational UI.

Image Segmentation Mobile App | Flutter, TensorFlow Lite, Dart

2024

- Built a mobile app performing live semantic image segmentation using on-device ML inference with 3 deep learning models (DeepLab V3, MobileNet UNet, MobileNet BiSeNet). Clean architecture; showcased at Computer Vision Projects Expo 2024.

EDUCATION

Singapore Institute of Technology

BEng (Hons) in Information & Communications Technology — Software Engineering | Aug 2022 – Aug 2026

Ngee Ann Polytechnic

Diploma in Electronics & Computer Engineering | Apr 2016 – Mar 2020

Director's List (2017/2018) | Certificate of Academic Achievement (2020)

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Dart, HTML/CSS, SQL

Frontend: React.js, Next.js, React Native (Expo), Flutter, Tailwind CSS, shadcn/ui

Backend: Node.js, NestJS, SAP CAP, Cloudflare Workers/Durable Objects, REST APIs, Vercel AI SDK

AI / LLM: Claude API, Model Context Protocol (MCP), Tool-use / Agent Orchestration, Prompt Engineering, Mastra

Data: PostgreSQL, Supabase, MongoDB, Firebase, Drizzle ORM, Cloudflare KV/Vectorize

DevOps: Git, GitHub Actions

Auth: JWT, OAuth 2.0, SSO, SAP XSUAA, Supabase Auth

Practices: Clean Architecture, OOP, Agile/Scrum, Code Review, Automated Testing